POWER ELECTRONICS

Power Switches

Introduction

- Linear amplifier can deliver hundreds of watt to the load, at low distortion
- Dissipates half of the power provided by the dc supply
- Converts dc voltage into heat
- Can be replaced with a **switch** (distortion about 0.5%)
- When switch is off – no power dissipation
- When switch is on – low voltage drop
- 90% of power delivered to the load
- Much efficient and accurate

Low Side and High Side

- Switch circuit common
  - Allowing load to float
  - Low side switch

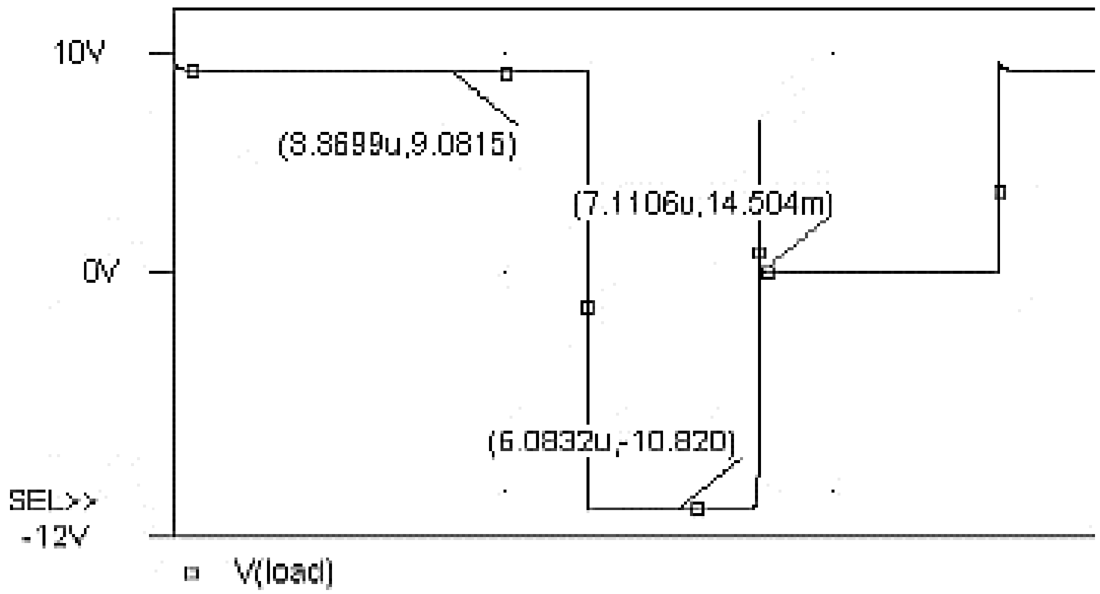
- Switch the power on and off
  - High side switch

Objectives

- To get familiar with switching characteristics of diodes and transistors, defining key parameters
- Study diode, Schottky diode, BJT transistor, MOSFET transistor as switch
- Familiar with the design and operation of low side and high side switch

Switching Characteristics, Diode

- Potential barrier, forward bias, reverse bias, Diode 1N4002
- For 60 Hz
- For higher frequencies like 100 kHz
- First problem:
  - Only 9 out of 10 V delivered to the load
  - 10% is dropped by the diode
  - Depending on diode, the drop may be 2 V
  - 2 V out of 160 V_P is insignificant (line powered app)
  - 2 V out of 5 V is too much (logic circuit)
- Second problem:
  - Diode doesn't turn off during negative half
  - Diode conducts in two directions



Switching Characteristics, Diode

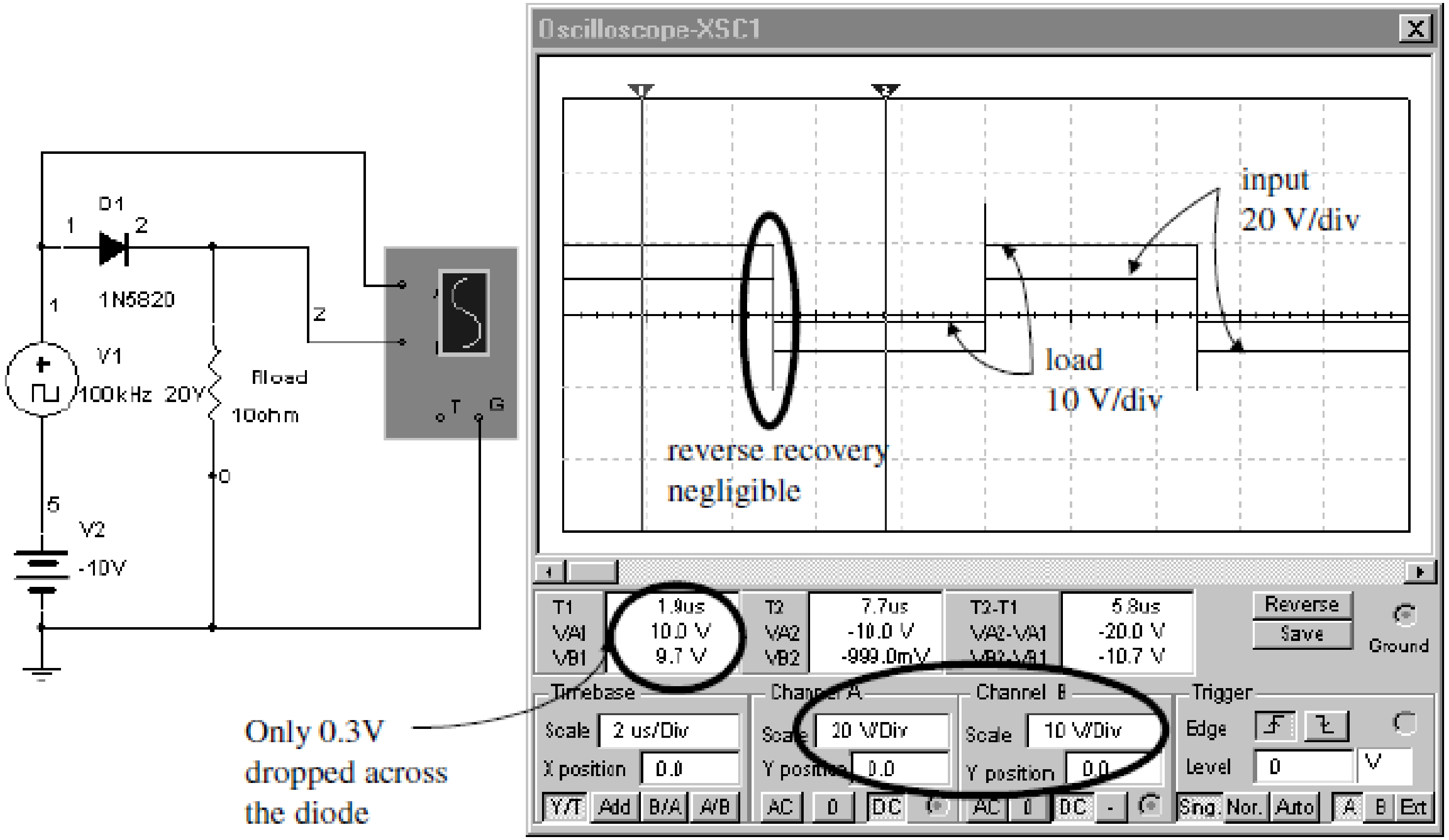
- Depletion region characteristics
- **Reverse Recovery Time ($\boldsymbol{t_{rr}}$)**
- For 60 Hz this time is a small part of the signal, no effect is observed
- At 100 kHz, significant part of the negative cycle is delivered to the load
- During t_{rr} the diode is not rectifying
- Select a diode whose $t_{rr} \ll t_{signal\,off}$

Schottky Diode

- No traditional pn junction
- Metal contact attached to $n-$ type semiconductor
- No $p-$ type material available
- By eliminating pn junction, both problems of traditional diode are solved
- Negligible voltage drop (on state)
- Negligible reverse recovery time

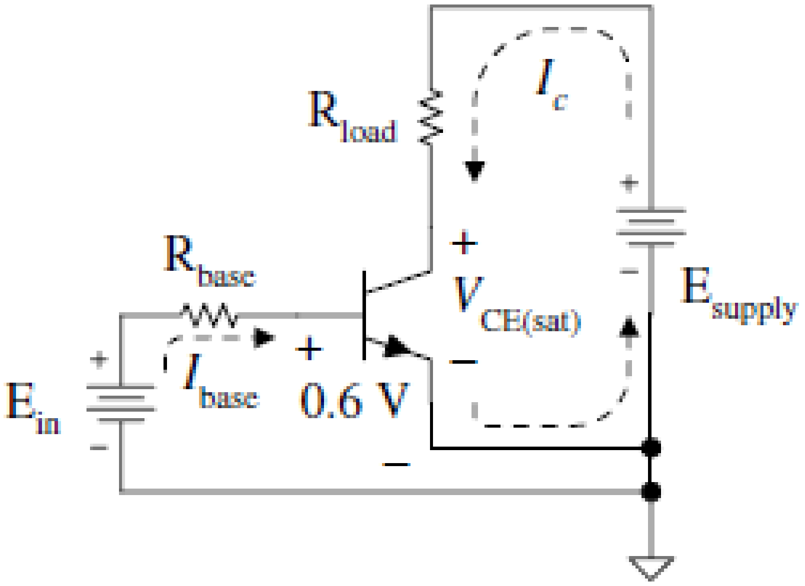
Schottky Diode

- 0.3 V drop
- Negligible reverse recovery
- **Problem with Schottky Diode:**
- Small **Reverse Voltage Breakdown**
- Usually less than 60 V
- **PIV** ratings



Switching Characteristics, Transistors

- Figure shows npn or pnp transistor?
- The base current depends on positive E_{in}
- $I_{base} = \frac{E_{in} - 0.6}{R_{base}}$
- Collector current depends on base current
- $I_c = \beta I_{base}$
- For power transistor β may be as low as 20
- For switching, transistor must be turned on, passing all current
- $I_c = \frac{E_{supply} - V_{CE(sat)}}{R_{load}}$
- Where 0.2 V $< V_{CE(sat)} <$ 1.0 V
- $V_{CE(sat)}$ depends on transistor, I_c and temperature



Switching Characteristics, Transistors

- Higher the collector current, larger the voltage across the transistor switch
- To ensure the transistor is on as hard as possible, base current should be higher than the required value
- $I_{base} > \frac{I_c}{\beta}$

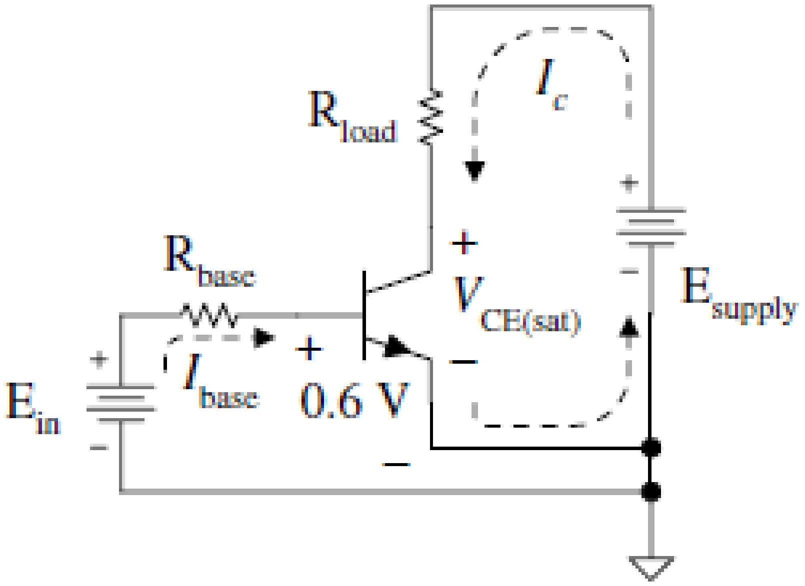
Example

- For the circuit shown in figure, $E_{supply} = 28\ \mathrm{V_{dc}}$, $E_{in} = 5\ \mathrm{V_{dc}}$, $R_{load} = 15\ \Omega$

$V_{CE(sat)} = 0.8$ V, $\beta = 20$

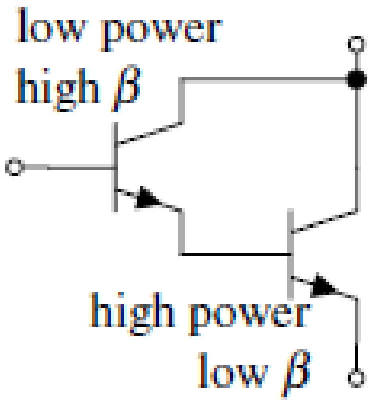
Calculate the following:

I_C, P_{load}, P_Q, I_{base}, R_{base}



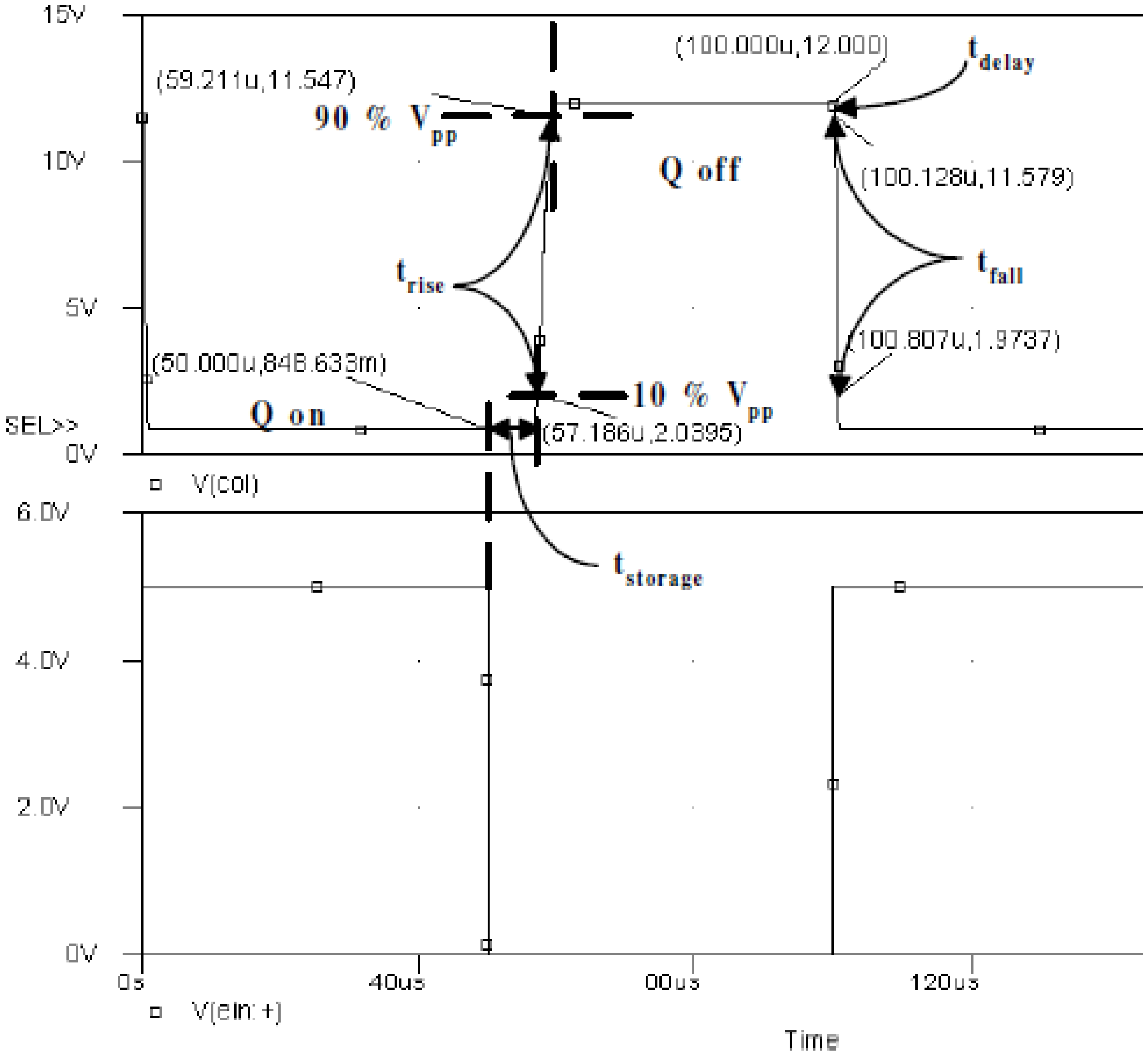
Key Points in Previous Example

- Almost 50 W is delivered to load while the power dissipation in transistor is only 1.5 W
- Much better than delivering power with push-pull amplifier
- 90 mA base current is so large (because β is small)
- By increasing β, base drive requirements drop
- High β and high collector current is possible with **Darlington Pair** configuration
- Single package with three terminals; base, emitter and collector



High Frequency Testing

- $E_{supply} = 12$ V
- 10 kHz, 50% duty cycle, 5 V input
- When on, $V_{col} = 0.849$ V
- When input goes low, turns off and $V_{col} = 12$ V
- **Storage Time** (transistor remains on)
  - Middle of input edge to 10% rise point in V_{col}
- **Rise Time** (V_{col} rises from 10% to 90%)
  - Depends on load resistance and capacitance
- **Delay Time** (to turn the transistor on)
  - Middle of rising edge to 90% point in V_{col}
- **Fall Time** (V_{col} falls from 90% to 10%)
  - all capacitors are discharged



Problems with Darlington Pair

- In previous circuit, the frequency is 10 kHz, produces high pitch whine
- Interference (too much annoying)
- Solution: run the circuit at 100 kHz
  - Storage time becomes greater than pulse width
  - Transistor never goes off
- Conclusion: Darlington Pair can not switch above the audio range

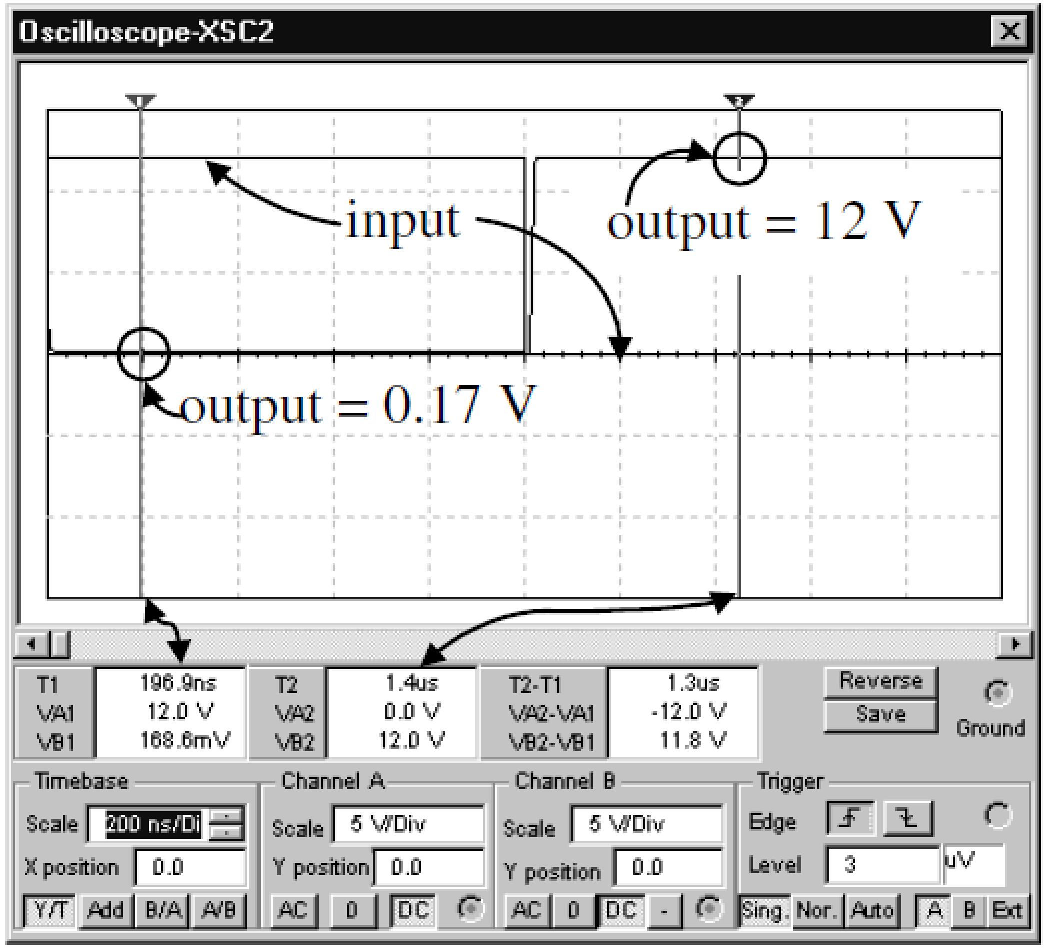
Enhancement Mode MOSFET

- Can be used as switch
- Larger the gate-to-source voltage, larger the channel between the source and drain
- Therefore, lower the channel resistance
- To turn the transistor hard on $10\ \mathrm{V} < V_{GS} < 20\ \mathrm{V}$

Comparison with Darlington Pair

- Comparable to Darlington pair discussed before but it operates 50 times faster
- 0.17 V instead of 0.89 V in *on* state
- 0.33 W wastes only, four times better
- Little storage time
- Negligible rise and fall time

	Darlington Pair	MOSFET
Storage Time	7.19 μs	9.2 ns
Rise Time	2.03 μs	8.0 ns
Delay Time	128 ns	0.9 ns
Fall Time	679 ns	2.6 ns

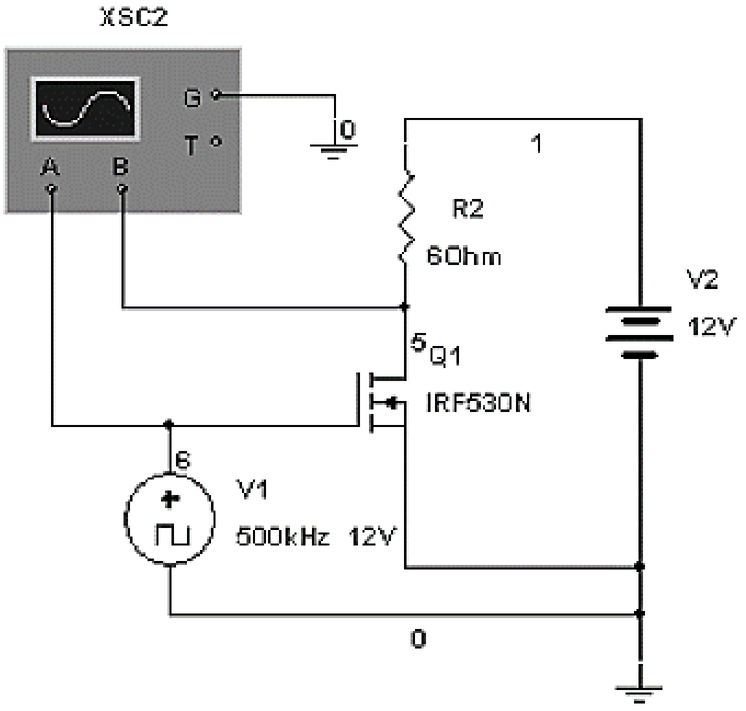


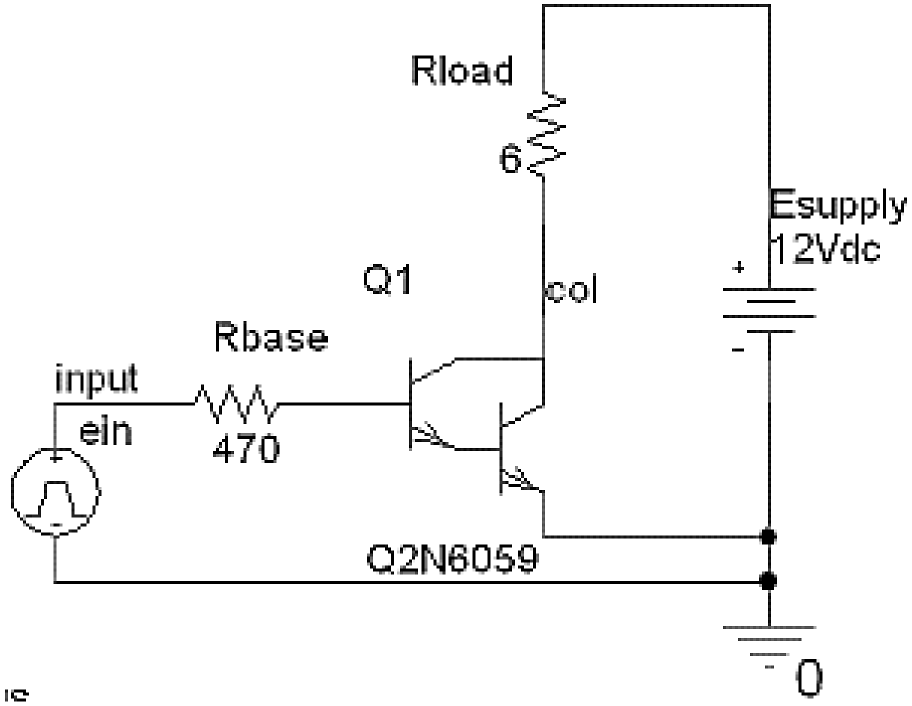
Parallel MOSFET Switches

- Drain to drain to drain…, source to source to source…, gate to gate to gate…
- R_{on} reduces from 0.11 Ω to 0.014 Ω in 8 parallel MOSFET
- 0.33 W reduces to 0.056 W
- Transistor stays cooler
- Less cost as compared to buy a heat sink for single transistor
- Less space required as compared to single transistor with heat sink

Low Side Switches

- Placed between load and circuit common,
- The other end of the load is pulled up by the supply





BJT as Low Side Switch

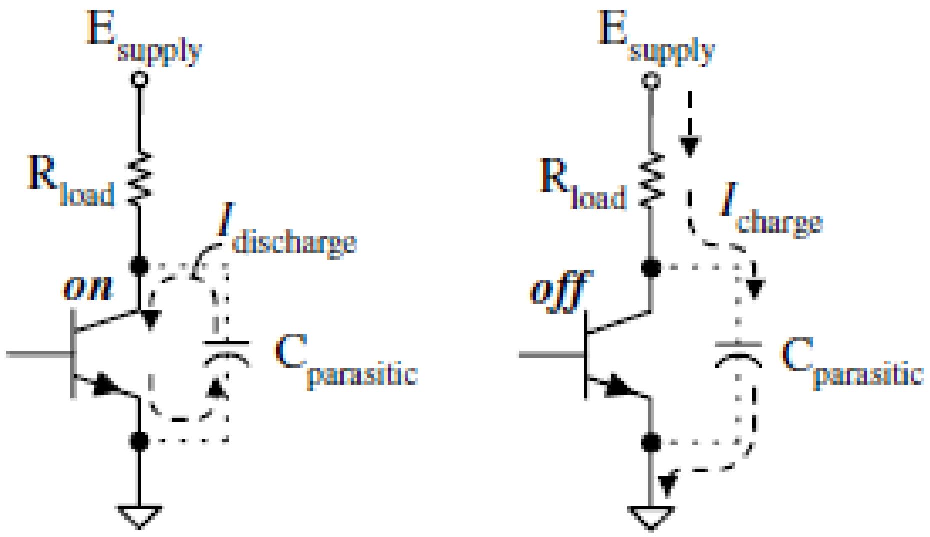
- BJT is a poor high speed, high current switch
- For applications that require 100 mA current or less, BJT works well
- Preferable to MOSFET in some cases
- Hard on from a supply voltage of 3.3 V to 5 V
- Smaller and lighter circuit board with BJT as compared to MOSFET
- For hard on BJT, $V_{BE} = 0.9$ V, $V_{CE} = 0.3$ V and $I_C = 50$ mA

Example

- Design a low side switch using 2N3904, a supply voltage of 28 V dc with a rectangular pulse input of 0 V to 2.4 V, 100 kHz. Set collector current to 50 mA peak. Use the smallest practical load resistance. $\beta = 100$
- Calculate P_{load} and P_Q

Parasitic Capacitance Effect

- Effect on rise and fall time
- $t_{rise\ RC} = 2.2R_{load}C_{parasitic}$



Example

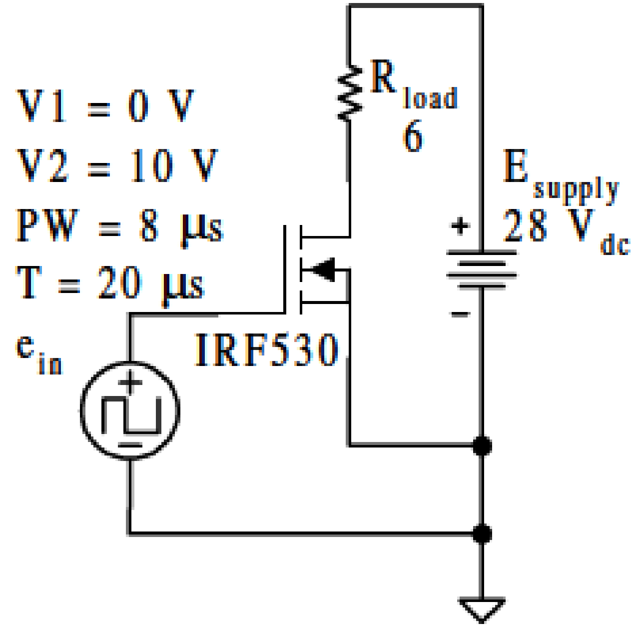
- Determine the rise time for $C = 640$ pF, $R_{load} = 560\ \Omega$, and for $C = 640$ pF and $R_{load} = 3.3\ \text{k}\Omega$ in the previous example.

MOSFET as Low Side Switch

- Switch several amps of current or more
- For speed less than 100 ns, enhancement mode MOSFET can be used
- $hard\ on$: $10\ \mathrm{V_p} \leq V_{GS} \leq 20\ \mathrm{V_p}$
- Opening resistance less than 0.11 Ω at 25°C
- Resistance increases when MOSFET heats up

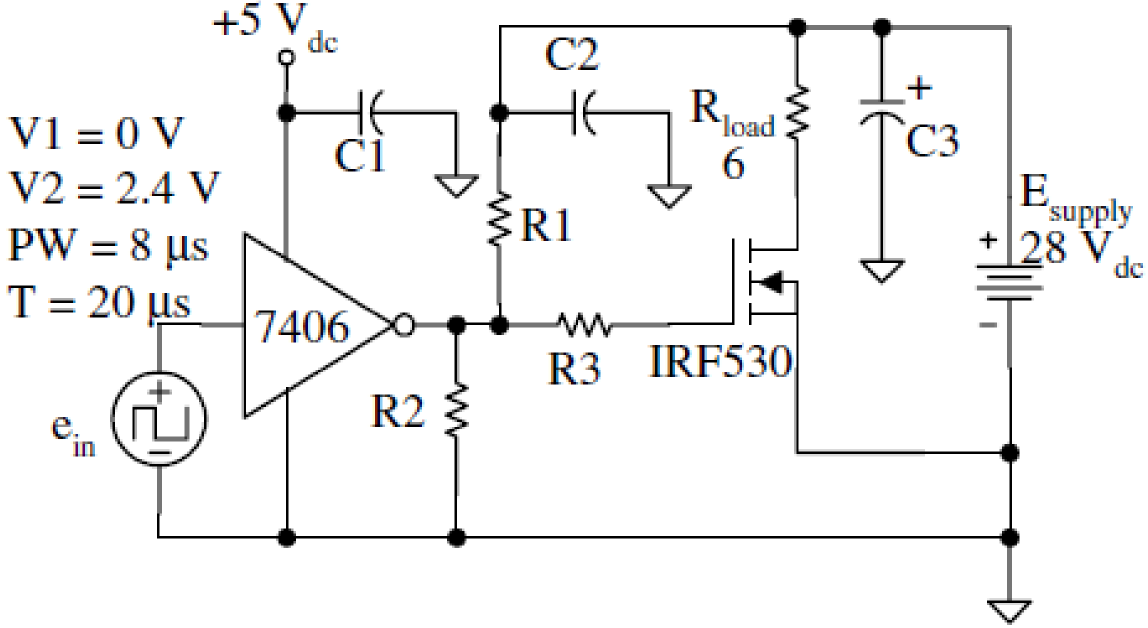
Example

- For the circuit shown in figure, the open resistance at 175 °C is scaled up by the factor of 2.6 over its resistance at 25 °C. Calculate $R_{on\,worstcase}$, $i_{p\,load}$, P_{load}, P_Q, T_J with $T_A = 40°C$ and no heat sink ($\Theta_{JA} = 62\ \frac{°C}{W}$)



Improved Circuit

- The current 4.45 A_P is far more than 7406 logic gate or the 2N3904 BJT can handle
- Less than half of the maximum of IRF530
- Input logic level comes from logic circuit or microprocessor = 2.4 V only
- 7406 to increase input to 10 to 20 V
- R_1 and R_2 provides voltage divider
- Capacitors are for decoupling the power circuits
- R_3 dampens the parasitic capacitance effect

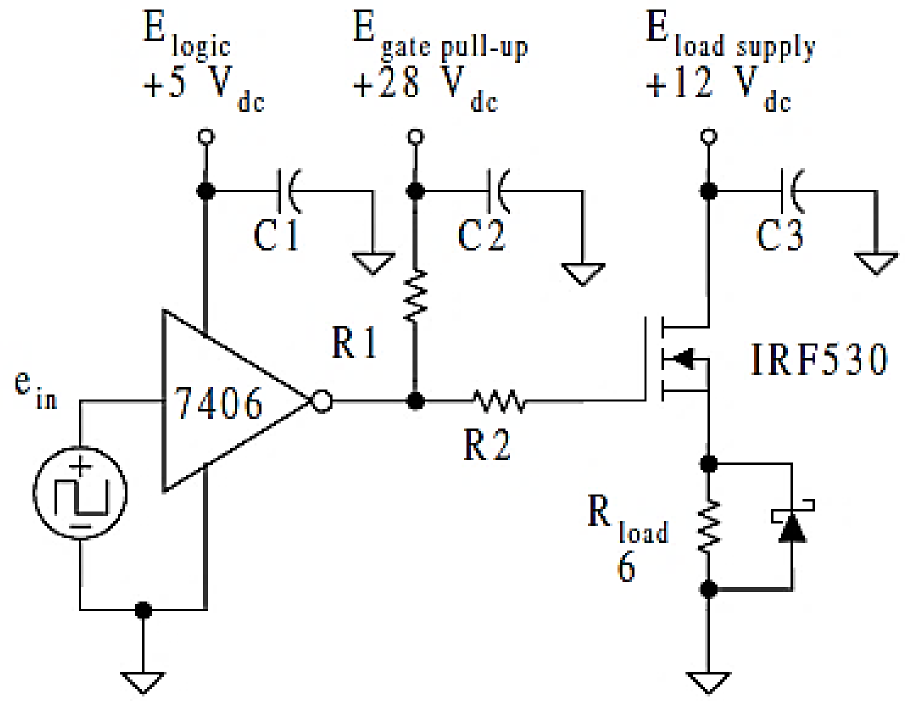


High Side Switch

- Many loads must be tied to common
- In this case, switch removes the voltage from the supply
- Switch sits on the **high side** of the load
- How to make high side switch? Your thoughts?

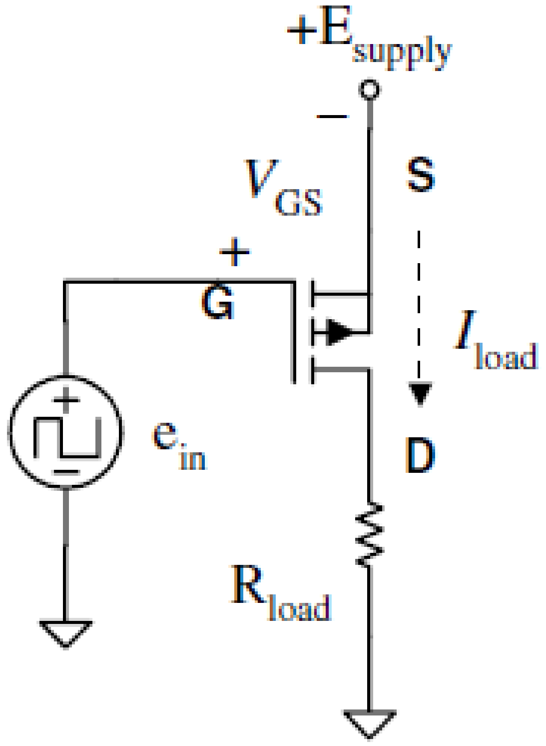
n-Channel MOSFET as High Side Switch

- Voltage divider removed from the MOSFET’s gate
- When input is high, 7406 outputs a low, turning the switch off
- When input goes low, 7406 output transistor goes on, the gate is pulled up to 28 V dc and the switch turns on
- When MOSFET is on, source voltage increases too
- Gate at +28 V, source at +12 V
- $V_{GS} = 28 - 12 = 16$ V



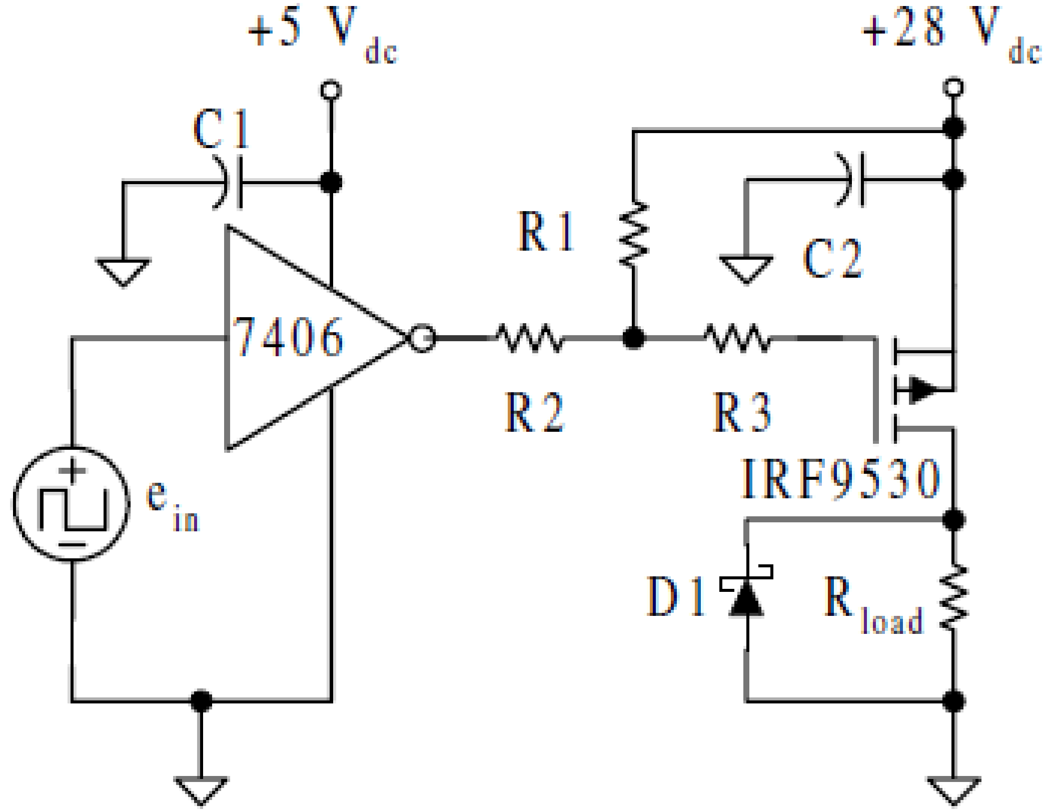
p-Channel MOSFET as High Side Switch

- Slower than n-channel MOSFET
- Can be used as high side switch
- $V_{GS} = V_G - V_S$



p-Channel MOSFET as High Side Switch

- Practical circuit as n-channel low side switch
- R_1 and R_2 should be selected as per current requirements
- 7406 can sinks up to 40 $\mathrm{mA_P}$ at output of 0.7 $\mathrm{V_P}$
- $8 \leq V_G \leq 18$

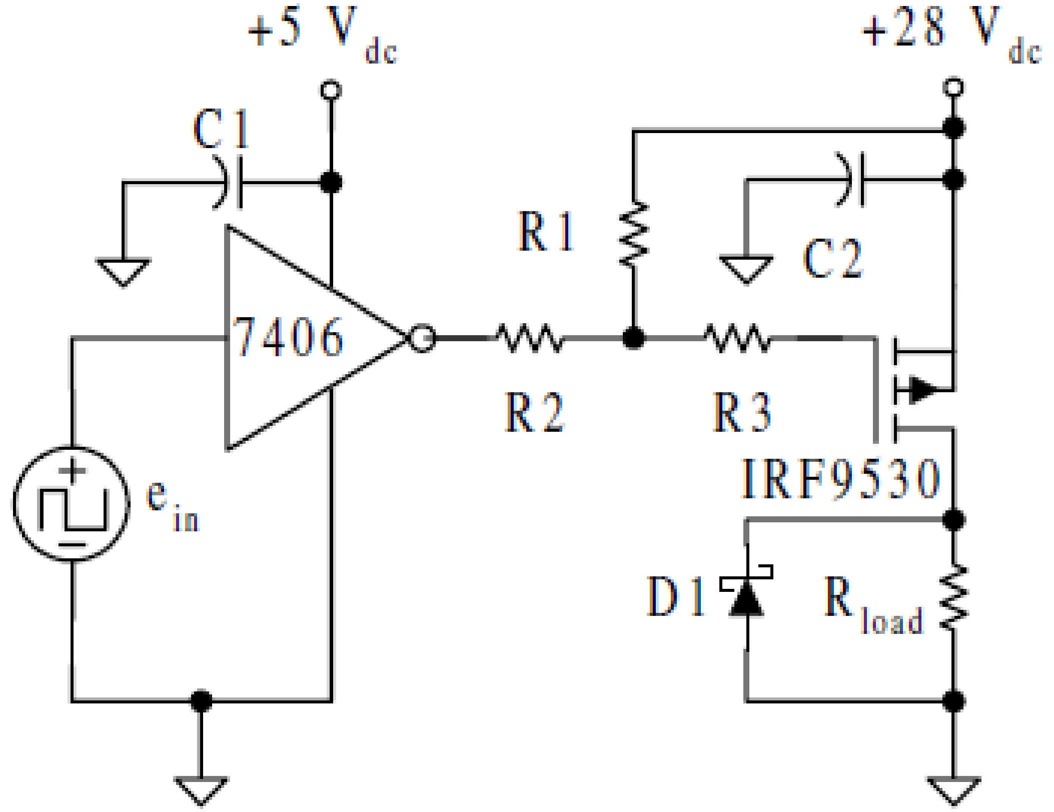


Example

- Design a p-channel high side switch
- For $R_{load} = 12\ \Omega$ determine voltage at each node when $e_{in} = 0.8\ V_P$ and when $e_{in} = 2.4\ V_P$
- Calculate v_{load} if IRF9530 on resistance is 0.57
- When e_{in} is a 30 kHz, 80% duty cycle, calculate:

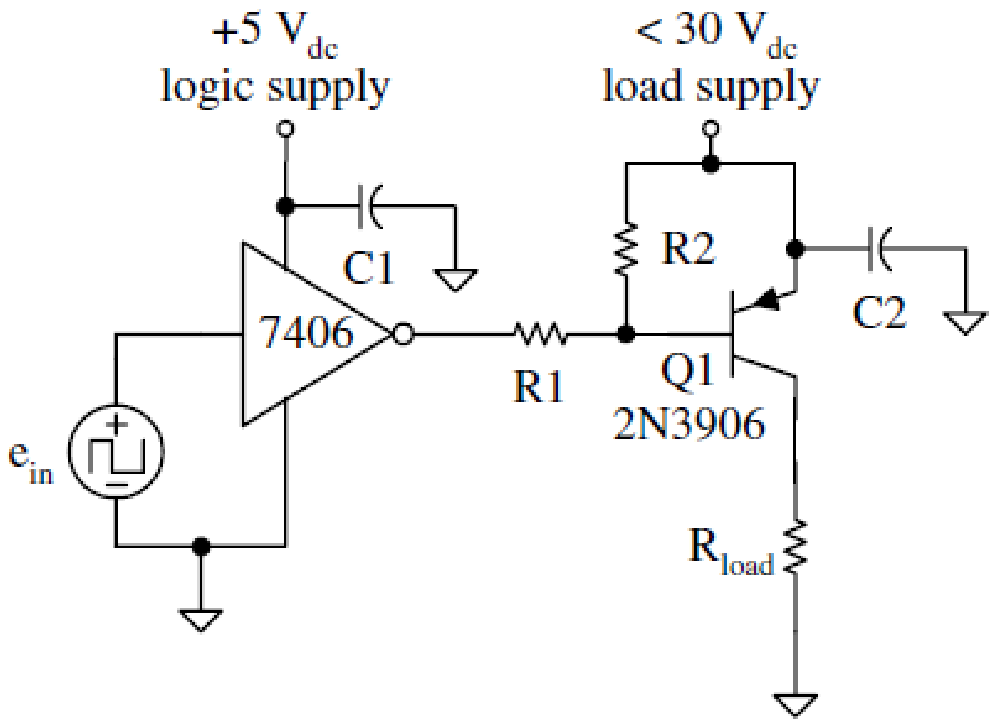
P_{load}, P_Q and Θ_{SA} if $T_{J\,max} = 150\ °\text{C}$ and $T_A = 50\ °\text{C}$

$\Theta_{JC} = 1.7\ °\text{C/W}$ and $\Theta_{CS} = 2\ °\text{C/W}$ with mica wafer



pnp High Side Switch

- For 100 mA peak of load current, or if switch speed is not critical, pnp BJT or Darlington pair can be used
- $i_{load} = ?$
- $i_{base} = ?$



Reference

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